

MeetingMind — Product & Architecture Overview

What It Is

MeetingMind is a self-hosted AI meeting intelligence platform that accepts any audio or video recording — regardless of length — and returns a structured, analyst-quality intelligence report. There is no practical file size or meeting duration limit. A 3-hour board meeting, a 90-minute client call, or a 6-hour conference session are all processed identically, automatically, and without user intervention beyond uploading the file.

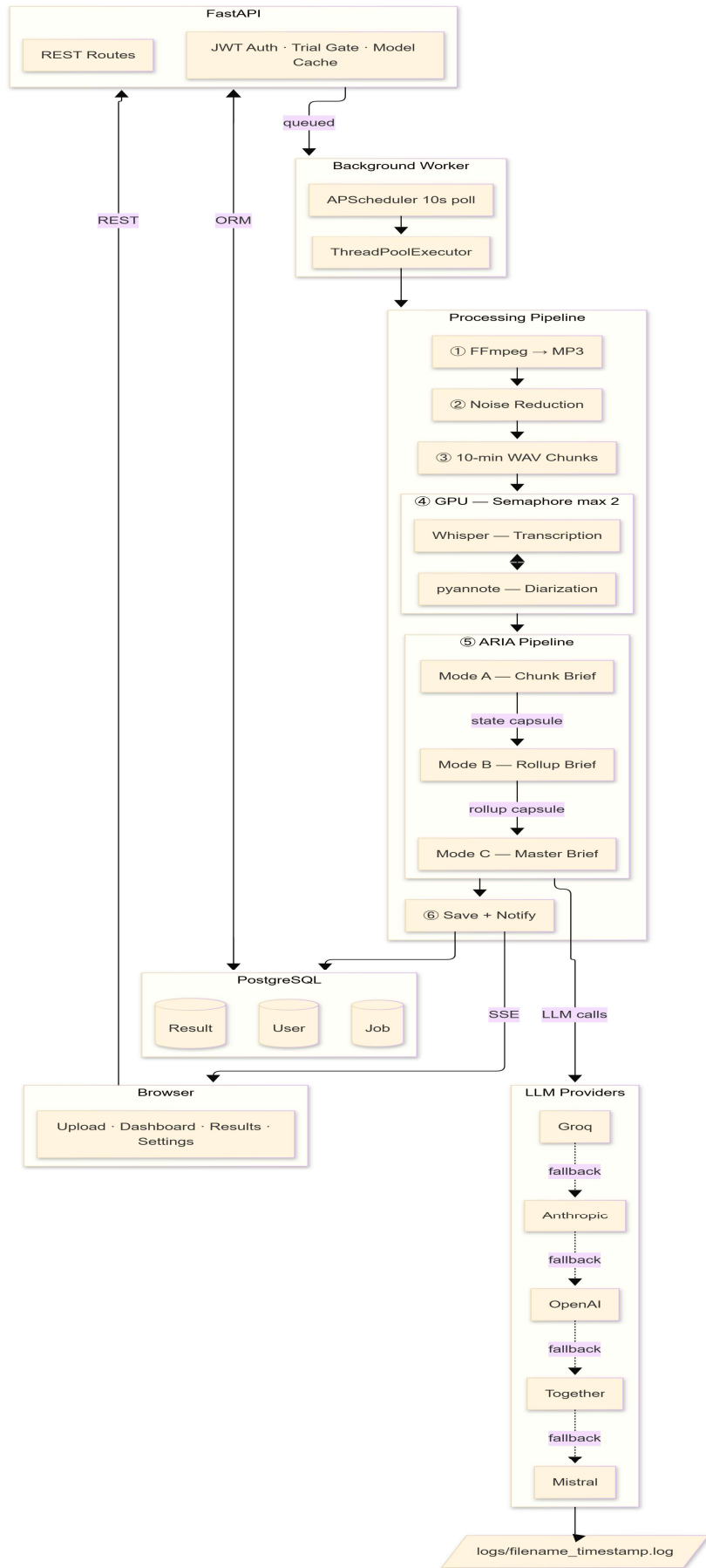
Architecture

The system is composed of five layers that work in sequence:

- **Browser (Frontend)** — A static vanilla JS interface for uploading files, viewing the dashboard, browsing results, and managing settings. Communicates with the backend over REST and polls the database for job status updates.
 - **FastAPI Application** — Handles all REST routes (upload, analyse, results, status, user settings, model listing). Enforces JWT authentication, a 3-job free trial gate, and caches available models per provider for 7 days.
 - **PostgreSQL** — Stores three core tables: User (API keys, provider/model preference), Job (status, metadata, duration), and Result (transcript, summary JSON).
 - **Background Worker** — An APScheduler polls for queued jobs every 10 seconds and dispatches each to a ThreadPoolExecutor thread, so the HTTP layer never blocks on processing.
-

Processing Pipeline — Each job passes through six stages:

- ① **Format Conversion** — FFmpeg converts any video/audio format to MP3 (192kbps, 44.1kHz)
 - ② **Noise Reduction** — FFmpeg DSP filters (highpass, FFT denoising, non-local means, loudnorm) controlled by NOISE_REDUCTION_LEVEL env var (low / medium / high)
 - ③ **Chunking** — FFmpeg splits the audio into 10-minute WAV chunks (16kHz, mono, PCM-16)
 - ④ **GPU Processing** — Whisper (faster-whisper on GPU, openai-whisper CPU fallback) transcribes each chunk; pyannote.audio 3.1 runs speaker diarization in parallel. A semaphore limits concurrency to 2 chunks at a time to prevent CUDA OOM on low-VRAM GPUs.
 - ⑤ **ARIA Pipeline** — Three-mode hierarchical LLM summarization:
Mode A: One LLM call per chunk, producing a brief + state capsule
Mode B: One LLM call per 4–6 chunks, rolling up briefs into a batch narrative
Mode C: Two LLM calls producing an overall summary JSON and the full 11-section intelligence report (Executive Summary, Key Topics, Decisions, Action Items, Sentiment, Glossary, and more)
 - ⑥ **Save + Notify** — Transcript and summary JSON are written to PostgreSQL; the client is notified via Server-Sent Events; temp files are cleaned up.
- LLM Providers** — A unified fallback chain: Groq → Anthropic → OpenAI → Together AI → Mistral. Users bring their own API keys. Every LLM call is logged to a per-job file (logs/filename_timestamp.log) with provider, model, token counts (prompt / output / total), latency, and full input/output.



Key Advantages

- No duration limit — The hierarchical ARIA pipeline (chunk → rollup → master) means a 6-hour meeting uses the same flow as a 30-minute one; the context window never overflows.
- Noise resilience — FFmpeg DSP preprocessing combined with tuned Whisper parameters (condition_on_previous_text=False) dramatically improves accuracy on real-world noisy recordings.
- Speaker-aware transcripts — pyannote 3.1 diarization labels every segment by speaker, enabling ARIA to attribute decisions and action items to specific people.
- ARIA state continuity — A structured state capsule chains chunk-to-chunk, preserving speaker identities, open topics, decisions, and action items across the full session — no context lost at chunk boundaries.
- Multi-provider LLM — 5 provider support with automatic fallback; users bring their own keys; model list fetched and cached live from each provider.
- Complete observability — Every LLM call logged with exact token counts, latency, provider, model, and full input/output — per job, per file, timestamped.
- GPU optimised — int8_float16 compute type, semaphore-limited parallel chunk processing, CUDA cache cleared per chunk — designed for consumer GPUs (4GB VRAM).
- PDF export — One-click print-to-PDF with full report expansion, correct print colours, Key Topics cards, and segment briefs — no external library required.
- Trial + subscription gating — 3 free uploads enforced at upload time; subscribed users get unlimited processing; both paths tracked in the database.
- Zero timeout risk — Jobs are queued in PostgreSQL, picked up by APScheduler every 10 seconds, and processed in a thread pool — the HTTP layer never waits for processing.

Tech Stack

Layer	Technology
API	FastAPI + Uvicorn
Database	PostgreSQL + SQLAlchemy ORM
Auth	JWT (HS256, 24h expiry) + bcrypt
Scheduler	APScheduler (BackgroundScheduler, 10s interval)
Audio processing	FFmpeg / ffprobe
Transcription	faster-whisper (GPU) / openai-whisper (CPU fallback)
Diarization	pyannote.audio 3.1
LLM providers	Groq, Anthropic, OpenAI, Together AI, Mistral
Frontend	Vanilla JS, HTML5, CSS3 (dark theme, no framework)
Deployment	Self-hosted, single-server, env-var configured

Processing At a Glance — 3-Hour Meeting

Stage	Operations	LLM Calls
Format + denoise + chunk	1 FFmpeg convert + 1 denoise + 1 split	0
Transcription + diarization	18 chunks × 2 models (parallel, max 2 GPU slots)	0
ARIA Mode A — Chunk Briefs	18 sequential calls	18
ARIA Mode B — Rollup Briefs	4 batch calls (batch size 5)	4
ARIA Mode C — Master Brief	S0 summary + full report	2
Total		24 LLM calls